Multivariate Calculus: Partials, Total Derivatives, and the Chain Rule

Math Foundations — Node 13

First, an example

Before the abstract definitions, hold a concrete picture in mind. Let

$$f(x, y) = x^2 y + \sin(xy).$$

Read aloud: f of x and y equals x squared times y plus sine of x y. Holding y fixed and differentiating in x, then holding x fixed and differentiating in y:

$$\frac{\partial f}{\partial x} = 2xy + y\cos(xy), \qquad \frac{\partial f}{\partial y} = x^2 + x\cos(xy).$$

Read aloud: partial f by partial x equals two x y plus y cosine of x y; partial f by partial y equals x squared plus x cosine of x y. At $(x, y) = (1, 2)$ this gives $\partial_x f(1, 2) = 4 + 2\cos 2$ and $\partial_y f(1, 2) = 1 + \cos 2$, so the gradient is $\nabla f(1, 2) = (4 + 2\cos 2,\ 1 + \cos 2)$. Keep this concrete pair of numbers in mind as the abstract machinery unfolds.

1 Definitions

Throughout, $U \subset \mathbb{R}^n$ is open and $f : U \to \mathbb{R}$. We write $\|\cdot\|$ for the Euclidean norm and $e_1, \ldots, e_n$ for the standard basis.

Definition 1 (Partial derivative)**.** Let $x \in U$. The *i-th partial derivative* of f at x is

$$\frac{\partial f}{\partial x_i}(x) := \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h},$$

Read aloud: partial f by partial x sub i at x is defined as the limit, as h goes to zero, of f of x plus h e sub i, minus f of x, all over h. provided the limit exists. Equivalently, holding all coordinates fixed except the i-th and viewing f as a univariate function of x_i, the partial is its ordinary derivative.

Definition 2 (Gradient)**.** When all n partials of f at x exist, the *gradient* of f at x is

$$\nabla f(x) := \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_n}(x) \right) \in \mathbb{R}^n.$$

Read aloud: nabla f at x is defined as the tuple whose i-th entry is partial f by partial x sub i at x, an element of R to the n.

Definition 3 (Directional derivative)**.** For a unit vector $v \in \mathbb{R}^n$, the *directional derivative* of f at x in direction v is

$$D_v f(x) := \lim_{h \to 0} \frac{f(x + hv) - f(x)}{h}.$$

Read aloud: D sub v of f at x is defined as the limit, as h goes to zero, of f of x plus h v, minus f of x, all over h.

Definition 4 (Total / Fréchet derivative). f is *differentiable at* $x \in U$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}$ such that

$$\lim_{h \to 0} \frac{\|f(x+h) - f(x) - Lh\|}{\|h\|} = 0.$$

Read aloud: the limit, as h goes to zero, of the norm of f of x plus h minus f of x minus L h, divided by the norm of h, equals zero. If L exists it is unique; we write $Df(x) := L$. Identifying linear maps $\mathbb{R}^n \to \mathbb{R}$ with row vectors via the inner product, $Lh = \nabla f(x) \cdot h$.

Remark 5. Existence of all partials at x does *not* imply differentiability, nor even continuity. A standard counterexample is $f(x,y) = xy/(x^2+y^2)$ on the punctured plane, extended by $f(0,0) = 0$: both partials vanish at the origin, but along $y = x$ the function equals $1/2$, so it is discontinuous there.

2 From partials to the total derivative

Proposition 6 (Continuous partials imply differentiability). *If $\partial f/\partial x_i$ exist in an open neighborhood of x and are continuous at x, then f is differentiable at x with $Df(x)h = \nabla f(x) \cdot h$.*

A proof is in Rudin, Theorem 9.21. In ML we work almost exclusively with C^1 functions, so this proposition lets us identify differentiability with the gradient picture without further worry.

3 The chain rule

The central result of this node is the multivariate chain rule along a curve.

Theorem 7 (Multivariate chain rule, scalar codomain). *Let $U \subset \mathbb{R}^n$ be open, $f : U \to \mathbb{R}$ differentiable at $a \in U$, and $\gamma : I \to U$ a curve on an open interval $I \subset \mathbb{R}$ with $\gamma(t_0) = a$ and γ differentiable at t_0. Then $f \circ \gamma : I \to \mathbb{R}$ is differentiable at t_0 and*

$$(f \circ \gamma)'(t_0) \; = \; \nabla f(a) \cdot \gamma'(t_0) \; = \; \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a)\, \gamma_i'(t_0).$$

Read aloud: *the derivative of f composed with gamma at t-naught equals the gradient of f at a dotted with gamma-prime at t-naught, which equals the sum from i equals one to n of partial f by partial x sub i at a times gamma sub i prime at t-naught.*

Proof. Set $L := Df(a)$, identified with $\nabla f(a) \in \mathbb{R}^n$ via $Lw = \nabla f(a) \cdot w$. Differentiability of f at a means: for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\|w\| < \delta \implies \big|f(a+w) - f(a) - Lw\big| \le \varepsilon \, \|w\| \, . \tag{1}$$

Read aloud: norm of w less than delta implies the absolute value of f of a plus w minus f of a minus L w is at most epsilon times the norm of w. Equivalently, there is a function $\varphi : \mathbb{R}^n \to \mathbb{R}$ with $\varphi(w) \to 0$ as $w \to 0$ and

$$f(a+w) - f(a) = Lw + \varphi(w)\, \|w\| \, , \qquad \varphi(0) := 0. \tag{2}$$

Read aloud: f of a plus w minus f of a equals L w plus phi of w times the norm of w, with phi of zero defined to be zero.

Differentiability of γ at t_0 similarly gives: there is a function $\psi : I - t_0 \to \mathbb{R}^n$ with $\psi(s) \to 0$ as $s \to 0$ and

$$\gamma(t_0 + s) - \gamma(t_0) = s\,\gamma'(t_0) + s\,\psi(s), \qquad \psi(0) := 0. \tag{3}$$

Read aloud: gamma of t-naught plus s minus gamma of t-naught equals s times gamma-prime of t-naught plus s times psi of s, with psi of zero defined to be zero.

Set $w(s) := \gamma(t_0 + s) - a$. By (3), $w(s) = s(\gamma'(t_0) + \psi(s))$, so

$$\|w(s)\| \le |s|\big(\|\gamma'(t_0)\| + \|\psi(s)\|\big). \tag{4}$$

Read aloud: the norm of w of s is at most the absolute value of s times the quantity norm of gamma-prime of t-naught plus norm of psi of s. Pick s small enough that $\gamma(t_0 + s) \in U$ and $\|w(s)\| < \delta$ (possible since γ is continuous at t_0). Substituting $w = w(s)$ into (2):

$$f(\gamma(t_0 + s)) - f(a) = Lw(s) + \varphi(w(s))\,\|w(s)\|\,.$$

Read aloud: f of gamma of t-naught plus s minus f of a equals L applied to w of s, plus phi of w of s times the norm of w of s. By linearity of L and (3),

$$Lw(s) = L\big(s\gamma'(t_0) + s\psi(s)\big) = s\,L\gamma'(t_0) + s\,L\psi(s).$$

Read aloud: L applied to w of s equals L applied to s times gamma-prime of t-naught plus s times psi of s, which equals s times L of gamma-prime of t-naught plus s times L of psi of s. Therefore

$$\frac{f(\gamma(t_0 + s)) - f(\gamma(t_0))}{s} = L\gamma'(t_0) + L\psi(s) + \varphi(w(s)) \cdot \frac{\|w(s)\|}{s}. \tag{5}$$

Read aloud: the difference quotient, f of gamma of t-naught plus s minus f of gamma of t-naught, divided by s, equals L of gamma-prime of t-naught plus L of psi of s plus phi of w of s times the norm of w of s over s.

We send $s \to 0$ term by term. The first term $L\gamma'(t_0) = \nabla f(a) \cdot \gamma'(t_0)$ is the target. For the second, L is continuous (every linear map on $\mathbb{R}^n$ is), and $\psi(s) \to 0$, so $L\psi(s) \to 0$. For the third, $w(s) \to 0$ by (4), so $\varphi(w(s)) \to 0$, while $|\,\|w(s)\|\,/s|$ is bounded above by $\|\gamma'(t_0)\| + \|\psi(s)\|$, which stays bounded near $s = 0$; hence the product tends to 0.

Combining, the limit of the left-hand side of (5) exists and equals $\nabla f(a) \cdot \gamma'(t_0)$. That limit is by definition $(f \circ \gamma)'(t_0)$, completing the proof. □

Remark 8 (The directional-derivative corollary). Take $\gamma(t) = x + tv$ with $v \in \mathbb{R}^n$ a unit vector. Then $\gamma'(0) = v$, and Theorem 7 specializes to

$$D_v f(x) = (f \circ \gamma)'(0) = \nabla f(x) \cdot v.$$

Read aloud: D sub v of f at x equals the derivative of f composed with gamma at zero, which equals the gradient of f at x dotted with v. This identity is the basis of the Cauchy–Schwarz argument that the gradient points in the direction of steepest ascent: $D_v f(x) \le \|\nabla f(x)\|$ with equality at $v = \nabla f(x)/\,\|\nabla f(x)\|$.

4 Worked example

Let $f(x, y) = x^2 y + \sin(xy)$. Computing partials by holding the other variable fixed:

$$\frac{\partial f}{\partial x} = 2xy + y\cos(xy), \qquad \frac{\partial f}{\partial y} = x^2 + x\cos(xy).$$

Read aloud: partial f by partial x equals two x y plus y cosine of x y; partial f by partial y equals x squared plus x cosine of x y. At $(1,2)$: $\partial_x f = 4 + 2\cos 2$, $\partial_y f = 1 + \cos 2$. So $\nabla f(1,2) = (4 + 2\cos 2,\ 1 + \cos 2)$. For $v = (1,1)/\sqrt{2}$,

$$D_v f(1,2) = \frac{(4 + 2\cos 2) + (1 + \cos 2)}{\sqrt{2}} = \frac{5 + 3\cos 2}{\sqrt{2}}.$$

Read aloud: D sub v of f at one comma two equals four plus two cosine two plus one plus cosine two, all divided by root two, which equals five plus three cosine two over root two. The companion script `code.py` confirms these values numerically.